COMPUTER ORGANIZATION AND ARCHITECTURE (21CSS201T)

Unit – 1 : Session – 3 : SLO – 2 | ASCII and parity codes
Name: JATANPREET SINGH VIRDI JATANPREET SINGH VIRDI | Reg. No: RA2511003011374 | Solved Worksheet

MCQs

• 1. b — 7
• 2. b — 65–90
• 3. c — Detect errors
• 4. b — 0–31 and 127
• 5. d — Even parity

Two-Mark Questions

• 1. ASCII (American Standard Code for Information Interchange) is a 7-bit code assigning a unique number to each letter, digit and control symbol. It lets any two digital systems exchange text unambiguously, forming the basis of digital communication.
• 2. Even parity adds a bit so the TOTAL number of 1s (including parity) is even; odd parity makes the total odd. Both detect a single-bit error; they differ only in the target parity convention.

Scenario-Based Long Answer

• Encoding: each character of the message is converted to its 7-bit ASCII code; a parity bit is computed and appended (MSB) to form an 8-bit frame; frames are transmitted over the noisy link.
• Verification: the receiver recounts the 1s in each received frame; if the parity does not match the agreed convention, the frame is flagged as erroneous and retransmission is requested.
• Example (even parity): letter 'A' = 65 = 1000001 (two 1s — already even) → parity bit 0 → sent as 01000001. If noise flips one bit to 01000011 (three 1s — odd), the receiver detects the parity violation and rejects the frame.

Match the Following

• ASCII — a) 7-bit code system
• Control characters — b) Range 0–31, 127
• Special characters — c) Used for punctuation
• Even parity — d) Total 1s made even
• Parity bit — e) Detects single-bit error

Fill in the Blanks

• 1. American Standard Code for Information Interchange 2. 65 3. odd 4. error 5. 7

Scenario-Based Small Activity

• 'B' = ASCII 66. 7-bit binary: 66 = 64 + 2 → 1000010.
• Count of 1s in 1000010 = 2 (already even), so the even-parity bit = 0.
• Final 8-bit code with parity as MSB: 0 1000010 = 01000010.

True or False

• 1. TRUE 2. TRUE 3. FALSE — Even parity makes the count even.
• 4. FALSE — Parity only detects single-bit errors; it cannot correct, nor detect all multi-bit errors. 5. TRUE